



Interpreting Probability Distributions

The likelihood of many events can be summarized by a table or histogram.

Interpreting Probability Distribution Tables

A summer camp program director is planning activities for the summer. The probability distribution table shows the probability, P , that the age of a randomly chosen camper is X .

X	10	11	12	13	14
P	0.15	0.3	0.2	0.25	0.1

For the first activity, the program director selects a camper at random to read announcements. What is $P(X = 13)$, the probability that the camper selected is 13 years old?

According to the probability distribution table, when $X = 13$, the probability is $P = 0.25$. So $P(X = 13) = 0.25$.

In other words, the probability of selecting a 13-year-old camper at random is 0.25, or 25%.

Suppose the program director wants to determine $P(X \geq 13)$, the probability that a randomly selected camper is at least 13 years old.

This problem can be solved by adding individual probabilities.

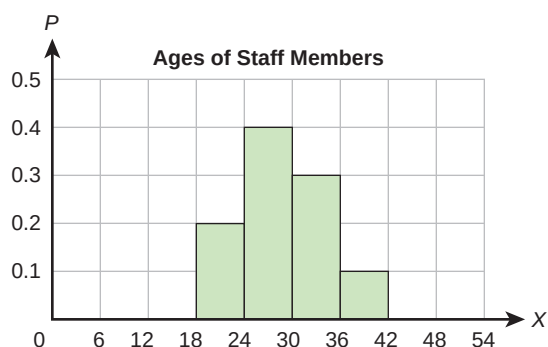
$$\begin{aligned}P(X \geq 13) &= P(X = 13) + P(X = 14) \\&= 0.25 + 0.1 \\&= 0.35\end{aligned}$$

There is a 35% probability of selecting a camper who is at least 13 years old.

Interpreting Probability Distribution Histograms

The program director now has to decide which staff members to assign to which activities. She is using the probability distribution histogram to help her make her decision.

In this histogram, X stands for the ages of individual staff members. P is the probability that if a single staff member is chosen at random, then the staff member will be that age.



► **Remember** A histogram is a graph showing the frequency distribution of data.

For the first activity, the program director selects a staff member at random to lead the songs at dinner. What is $P(18 \leq X < 24)$, the probability that the staff member selected is at least 18 years old and less than 24 years old?

The first bar of the histogram represents $18 \leq X < 24$. The height of this bar is at the value $P = 0.2$. So the probability of selecting a staff member who is at least 18 years old and less than 24 years old is 0.2, or 20%.

Probability distribution histograms can help determine more than just single entries. Suppose the program director wants to determine $P(X \geq 30)$, the probability that a randomly selected staff member is at least 30 years old.

This problem can be solved by adding individual probabilities.

$$\begin{aligned} P(X \geq 30) &= P(30 \leq X < 36) + P(36 \leq X < 42) \\ &= 0.3 + 0.1 \\ &= 0.4 \end{aligned}$$

There is a 40% probability of selecting a staff member who is at least 30 years old.